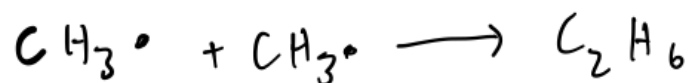


30 Gas phase kinetics

In the last lecture, we discussed how homogeneous gas phase reactions occur when gas molecules collide with each other. In this lecture, we will look deeper into the molecular phenomena that occur during gas phase chemistry.

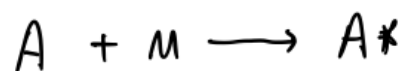
Collisions between gas species can result in a reaction, for example:



However, more often, collisions between gas molecules result in the formation of

a n active intermediate

For example:



here $*$ = active intermediate,
NOT species adsorbed to a catalyst

An active intermediate is a high energy molecule that reacts virtually as fast as it is formed

During the collision, kinetic energy is transferred into internal energy

where internal energy comprises electronic, vibrational, and rotational contributions

In other words, the energy is absorbed into the chemical bonds of the molecule

which causes perturbation to the electronic structure

Active intermediates formed via collision can include excited states, molecular re-arrangements, breaking/forming of bonds, etc

Due to their very short lifetimes, active intermediates can be modeled using the

QSSA

Pseudo steady state hypothesis / quasi steady state approximation
PSSH

The PSSH assumes

that a species is at steady state even if the system it's in is not. This means that the net rate of generation of the species = 0. $\frac{dC_A^*}{dt} = 0$.

An intermediate species is

Species that are formed during a chemical rxn that are not reactants or products

Let's do an example: The azomethane decomposition reaction is proposed to occur via the following mechanism:

1. $(\text{CH}_3)_2\text{N}_2 + (\text{CH}_3)_2\text{N}_2 \rightarrow (\text{CH}_3)_2\text{N}_2 + [(\text{CH}_3)_2\text{N}_2]^*$) we could write this as one
2. $[(\text{CH}_3)_2\text{N}_2]^* + (\text{CH}_3)_2\text{N}_2 \rightarrow (\text{CH}_3)_2\text{N}_2 + (\text{CH}_3)_2\text{N}_2$) reversible reaction or two irreversible
3. $[(\text{CH}_3)_2\text{N}_2]^* \rightarrow \text{C}_2\text{H}_6 + \text{N}_2$ ← irreversible rxn

What is a mechanism? a list of elementary steps that combine to form a chemical reaction

↓ denote Azo

What is happening in reaction 1? Why is $(\text{CH}_3)_2\text{N}_2$ both a reactant and a product??

two Azo molecules collide, forming one active intermediate, Azo^* . the other Azo acts more like a catalyst here than a reactant/product.

What does the * mean?

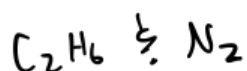
species is an active intermediate, likely excited state or molecular re-arrangement since there is no change in stoichiometry

What is happening in reaction 2?

reverse of rxn 1 $\rightarrow$ represents $Azox$ decaying back to unactivated AzO

What is happening in reaction 3?

active intermediate reacts (decomposes) into



What are the intermediate species in the mechanism?

overall rxn: $(CH_3)_2N_2 \rightarrow C_2H_6 + N_2$. $Azox$ is an intermediate species & also an active intermediate.

Write the rate law for the formation of ethane. Then, use the PSSH to eliminate the concentrations of all intermediate species from the rate law. What are the apparent reaction orders at high and low concentration of $(CH_3)_2N_2$?

$$r_{C_2H_6} = k_3 C_{Azox}$$

Problem: Can't measure C_{Azox} b/c $Azox$ is highly reactive. So, use PSSH to approximate C_{Azox} .

$$\text{net rate of generation of } Azox = 0$$

$$0 = k_1 C_{A_{20}}^2 - k_2 C_{A_{20}*} C_{A_{20}} - k_3 C_{A_{20}*}$$

• solve this in terms of the concentrations of reactant
 & product species & eliminate concentrations of all
 intermediate species

$$0 = k_1 C_{A_{20}}^2 - C_{A_{20}*} (k_2 C_{A_{20}} + k_3)$$

$$C_{A_{20}*} = \frac{k_1 C_{A_{20}}^2}{k_2 C_{A_{20}} + k_3}$$

$$r_{C_2H_6} = k_3 C_{A_{20}*} = \frac{k_3 k_1 C_{A_{20}}^2}{k_2 C_{A_{20}} + k_3}$$

in experiments, you can't measure all these parameters
 individually, instead you'll measure "lumped" parameters

@ high $C_{A_{20}}$: $k_2 C_{A_{20}} \gg k_3$

$$r_{C_2H_6} \approx k' C_{A_{20}} \quad \text{apparent 1st order}$$

@ low $C_{A_{20}}$: $k_3 \gg k_2 C_{A_{20}}$

$$r_{C_2H_6} \approx k' C_{A_{20}}^2 \quad \text{apparent 2nd order}$$

apparent $\rightarrow$ observed w/ the acknowledgment that
 some details are being obscured